2026 BROCK AI DAY · EXPLAINABLE AI / DEEP LEARNING

Are language models converging on shared structures?

Mechanistic interpretability, the linear representation hypothesis, and the platonic representation hypothesis

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1 · CONCEPTS ARE VECTORS: THE LINEAR REPRESENTATION HYPOTHESIS

A model stores each concept as a direction

WITH CREDIBILITY MARKERS

“Traditional vaccine development is a meticulous process, typically requiring 10–15 years…”

SAME CLAIM, NO GROUNDING

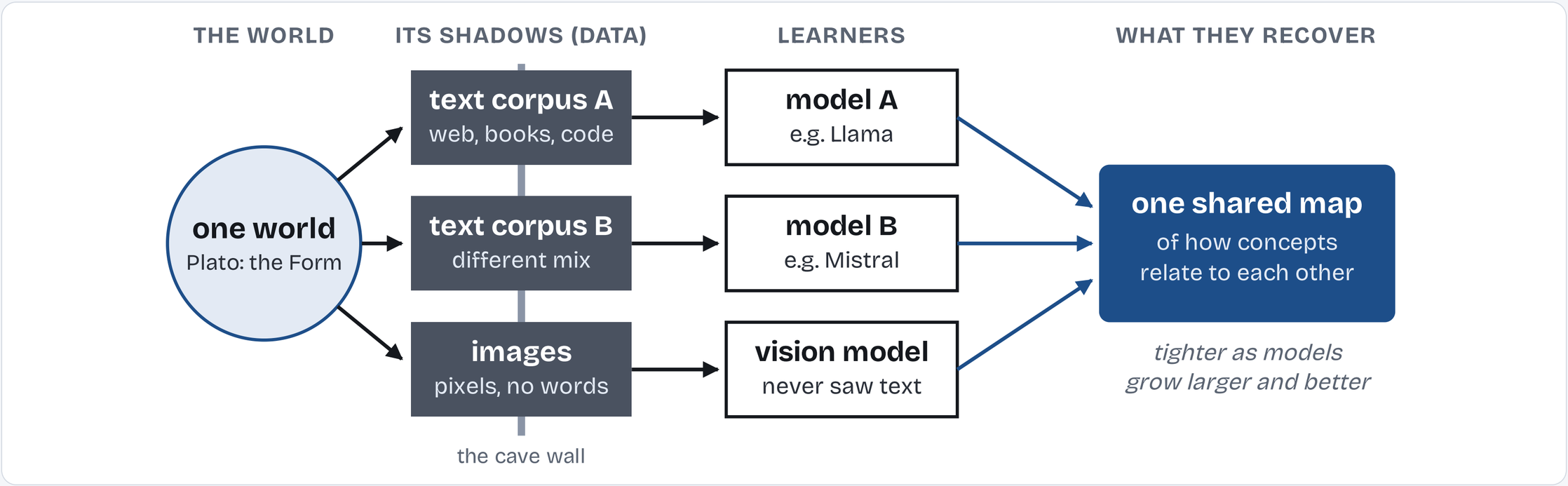
“It’s just common sense that a vaccine should take years to develop…”

mean(with) − mean(without) = the concept vector, at every layer

17 concepts · about 250 pairs each · written by 14 different LLMs

Park, K., Choe, Y. J., & Veitch, V. (2024). *The Linear Representation Hypothesis and the Geometry of Large Language Models.* ICML.

Different shadows, one world?



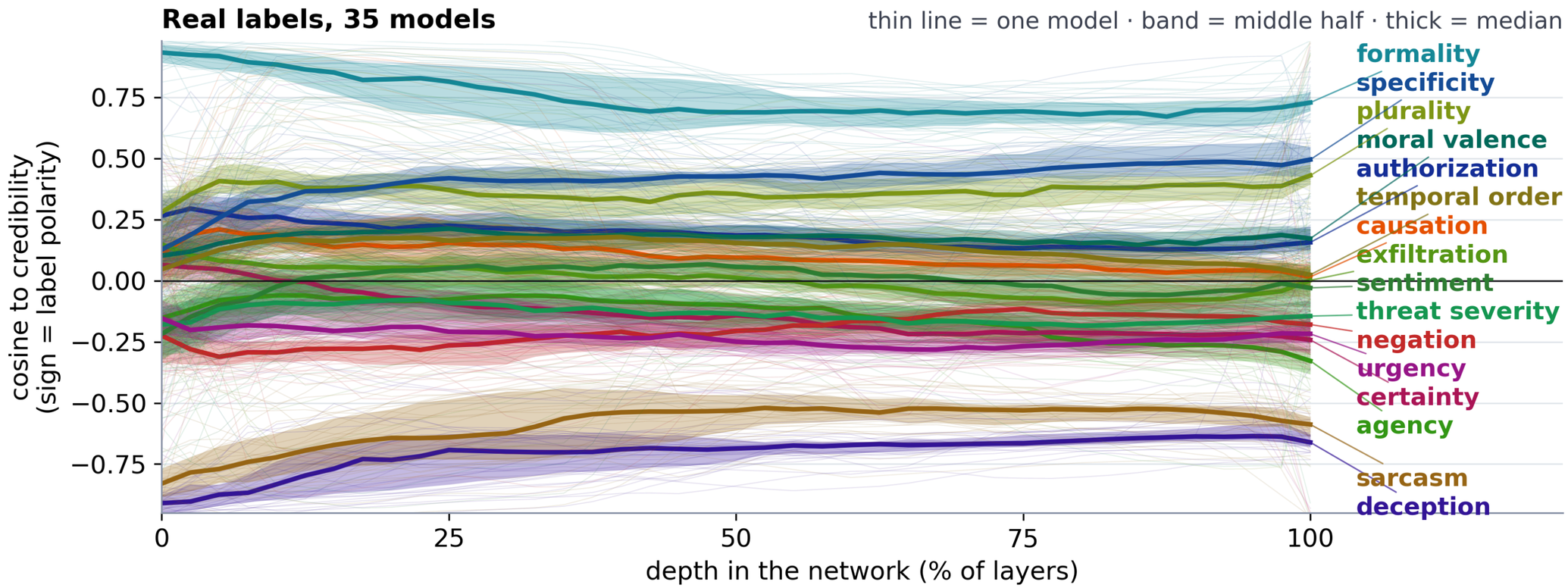
What converges is relations, not coordinates.

Prediction: models that never shared a weight should agree on how concepts relate to each other.

Huh, M., Cheung, B., Wang, T., & Isola, P. (2024). *The Platonic Representation Hypothesis.* ICML.

3 · OUR TEST: 35 MODELS, EVERY LAYER

What is credibility aligned with?



Each line is a concept's cosine to credibility across depth; bands show the spread over 35 models. At the final layer formality is closest in 29 of 35. Deception and sarcasm are nearly as strong with the opposite sign (sign = label polarity).

4 · A VIEW FROM OUTSIDE

Can the angles alone recognize a concept?

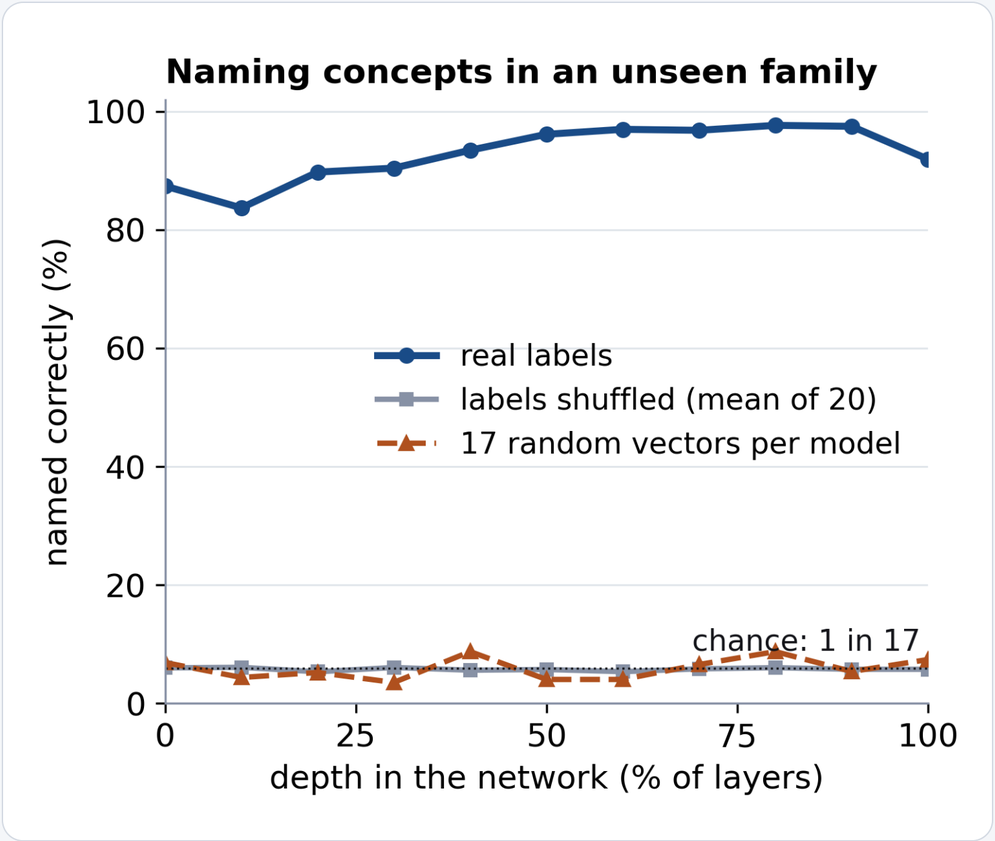
97%

recognized at mid-depth

6%

with shuffled labels or random vectors, which is chance

To look from the outside, set one whole family of models aside. Describe each of its concepts only by its angles to the others, and ask whether what the remaining families share is enough to recognize it.



WHAT IT MIGHT MEAN

The consolidation of meaning

Thirty-five models, built by different teams on different text, agree on how their concepts sit relative to one another. Each model's coordinates are its own; the angles between its concepts are not.

Is this the structure of language itself, or the structure of the world it describes?

Poster source, figure code and reading list: github.com/jamesrahenry/ai-day-2026